

Niklas Abraham

Altdorf, 71155

niklas.cv.abraham@gmail.com · +49 157 512 11717 · niklas-abraham.de

EDUCATION

University of Oxford, Department of Statistics

Recognised Student, Oxford Protein Informatics Group (OPIG)

Oxford, United Kingdom

Oct 2026 - Mar 2027

- Master's thesis research (MSc Machine Learning, University of Tübingen): **Over-conditioning in diffusion-based co-folding models: diagnosis and diversity recovery.**
- Supervised by Dr. Nele Quast and Prof. Charlotte M. Deane (OPIG), with Prof. Philipp Hennig (Tübingen) as formal supervisor.
- Investigating to what degree sequence conditioning causes mode collapse in co-folding models (AlphaFold3, Boltz-2, OpenFold3), where the denoiser returns a near-final structure in a single reverse step.
- Developing a region-resolved metric for mode collapse (scaffold vs. interface vs. ligand, across diffusion time), a benchmark of systems with known conformational heterogeneity, and inference- and train-time interventions to recover sample diversity without sacrificing physical validity.

University of Tübingen

Master of Science in Machine Learning

Tübingen, Germany

Oct 2025 - Oct 2027

- A four-semester English-language programme in the Faculty of Mathematics and Natural Sciences, embedded within one of Europe's leading ML research hubs through links with Max Planck Institutes and Cyber Valley.
- Core modules include Deep Learning, Statistical Machine Learning, and Probabilistic Inference and Learning.
- Intended focus: advanced generative modelling (score-based diffusion, flow-matching), non-convex optimisation in deep learning, and reinforcement learning for sequential decision-making, with applications in bioinformatics.
- Current grade average: 1.1 (33 Credits done)

University of Stuttgart

Bachelor of Science in Simulation Technology

Stuttgart, Germany

Oct 2022 - Sept 2025, Grade 1.9 (5th from 22)

- A bachelor programme focusing on mathematics, physics, computer science, applied and theoretical courses such as Calculus I-IV, numerical mathematics, machine learning and more.
- Elected focus on AI and bioinformatics.

German Scholarship Foundation (Studienstiftung des Deutschen Volkes)

Scholarship Holder

Germany

- Two-week summer programme in Switzerland - course work: "Sustainable development in developing countries"; visits to IFO and UN Geneva, reading and learning about new economic policies for developing countries.
- Three-week summer school in France (Montpellier) learning French while living with a host family and experiencing local culture.

Schönbuch-Gymnasium Holzgerlingen

German high-school diploma

Holzgerlingen, Germany

July 2022

- Graduated with prizes for excellence in STEM and economics - grade 1.0 (GPA 4.0).
- Founded the school's 3D-printing AG (student working group) and ran it for three years.

EXPERIENCE

Folaint

Co-Founder, CTO & Managing Director (Geschäftsführer)

Germany

2026 - present

- Co-founded an AI start-up building "a foundation your AI can reason on" (folaint.de); lead a 5-person team as Managing Director (Geschäftsführer) and own the full technical direction as CTO.
- Architect a knowledge-graph foundation on Neo4j that LLM agents reason over, together with the orchestration layer coordinating multi-agent LLM workflows on top of it.
- Own the end-to-end technical stack - GPU-accelerated model inference, Dockerised deployment, and the retrieval and reasoning pipelines - while setting the technical vision, research direction, and engineering priorities across the team.

Merck KGaA, Group Science & Technology Office

Research Intern (full-time, 3 months)

Darmstadt, Germany

Apr 2026 - Jun 2026

- **Enhanced Protein Function Prediction Through Multi-Mutant Interaction Modelling.**

- Conducted a comprehensive literature review and evaluation of methods for multi-site mutation property prediction, advancing beyond single-mutation approaches to capture epistatic interactions and non-additive effects in protein engineering.
- Synthesised critical insights on embedding strategies and feature engineering (ESM embeddings, structure-based representations, attention mechanisms) under the practical constraints of limited experimental datasets.
- Designed an implementation roadmap for novel prediction tools, establishing best practices for training-data composition, regularisation strategies, and uncertainty quantification in computational protein design.

Institute of Biochemistry and Technical Biochemistry

Researching Student

Stuttgart, Germany

2023 - 2025 December

- Developed a machine-learning framework integrating large protein language models (ESM-2, ESM-C) with curated biological databases (NCBI, CARD, UniProt) to predict β -lactamase resistance phenotypes directly from sequence data, demonstrating that embedding-level representations encode biochemically meaningful features.
- Introduced and benchmarked novel residue-level pooling strategies, including self-similarity weighted aggregation and biochemically informed positional weighting, achieving substantial performance gains over classical mean pooling and providing mechanistic insights into sequence–function relationships.
- Contributed to the development of the **Python Enzyme Engineering Database** (PyEED), leveraging Neo4J to build complex biological networks for proteins, DNA, and ontologies.
- Collaborated under Prof. Jürgen Pleiss to advance machine learning and bioinformatics methods for enzyme engineering, integrating graph databases and vector embeddings for protein and ontology analysis.

Levigo Solutions GmbH

Student Software Engineer

Holzgerlingen, Germany

Apr 2020 - Sept 2021

- Set up Docker/Kubernetes clusters utilising Elasticsearch.
- Developed custom plugins providing analytics for Docker containers.

Racing Team Fusion - “F1 in Schools”

Aerodynamics Engineer

Holzgerlingen, Germany

Sept 2018 - Sept 2019

- Achieved 2nd place at the World Championship in Abu Dhabi 2019 (best German result to date).
- Led vehicle development in the largest student engineering competition; coordinated the engineering team and applied principles of physics and mechanics.
- Designed testing methods, supported manufacturing, and gained expertise in CAD, Ansys Fluent, mesh settings, and technical drawing creation.
- Secured €30 000 in sponsorship from various German companies.

LEADERSHIP & ACTIVITIES

Merck M-Boltz Hackathon

Winner, Speed-Up Challenge

Darmstadt, Germany

October 2025

- Winner of the Speed-Up Challenge; developed a fast deterministic flow-matching framework (“Boltz Flow Matching”) enabling analytical conversion from score-based diffusion to flow equations for accelerated protein structure generation.
- Designed and implemented high-performance sampling pipelines integrating diffusion-model mathematics, improving generation speed and stability while preserving high degree of structural accuracy.

KiSS e.V. (local sport courses for children)

Climbing Trainer

Weil im Schönbuch, Germany

July 2022 - present

- Climbing trainer twice a week for two groups of ten children (ages 5–10).

University of Stuttgart - Faculty Committee

Member & Simulation Technology Student Speaker

Stuttgart, Germany

Aug 2023 - October 2025

- Elected representative for students of Simulation Technology.

Competitions and Activities

Germany

- Competitor - German National Artificial Intelligence Competition for High-School Students (May 2019).
- Junior Science Academy - course in technology and chemistry (July 2018).
- Founding roles in high-school 3D-printing and science clubs (2014 - 2022).
- “Jugend forscht” (physics) - 2nd place in Baden-Württemberg state finals (2017).

PUBLICATIONS

- **Converting diffusions to flows accelerates sampling and suggests over-conditioning of co-folding models on sequence.** Nele P. Quast, Niklas Abraham, Aaron Schöne, Fergus Imrie, Matthew I. J. Raybould, Yee Whye Teh, Charlotte Deane (Oxford Protein Informatics Group, University of Oxford).
ICLR 2026 Workshop on Learning Meaningful Representations of Life (LMRL), Poster (accepted).
We convert stochastic diffusion sampling to a deterministic flow process; pre-trained diffusion-based structure predictors as probability-flow ODEs give **equivalent FoldBench performance** with a **20× sampling speed-up** and show **strong conditioning on sequence and MSA**.
- **Geometric encoding of enzymatic mechanism in protein language model representations.** Jan Range, Niklas Abraham, Jürgen Pleiss (Institute of Biochemistry and Technical Biochemistry, University of Stuttgart).
Manuscript in preparation. Functionally critical residues form a **directional outlier subspace** in ESM-2 last-layer representations; **+14 percentage points** on EC sub-subclass classification vs. mean pooling, validated on ligand-binding sites (**median AUC 0.755**) and ProteinGym fitness assays **without alignments or structural data**.

SKILLS & INTERESTS

Technical: Python, PyTorch, Jax, CUDA, Docker, Neo4j, Java, C++, Git

AI systems: LLM agents, multi-agent orchestration, retrieval-augmented generation (RAG), knowledge graphs, protein language models

Hardware & CAD: Arduino, KiCad, Autodesk, Solid Edge

Languages: German (native), Polish (native), English (C1), French (B2)

Interests: Reading (all non-fiction books I can find, except for *Dune*); triathlon (next up: 70.3 Ironman); indoor climbing; debating politics and economics.